

VISHNU S

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PROFESSIONAL SUMMARY

Computer Science Engineering student specializing in Artificial Intelligence and Machine Learning with hands-on experience in Vision Transformers, LSTM, and CNN architectures. Experienced in building end-to-end AI systems integrated with React and FastAPI. Strong foundation in Data Structures, Algorithms, Probability, and scalable system design.

EXPERIENCE

Research Intern – Generative AI | NIT Calicut

Apr 2025 – Jun 2025

- Built a Vision Transformer (ViT)-based handwritten text recognition system for multilingual scripts, directly applicable to document understanding and RAG pipeline pre-processing.
- Designed a complete image preprocessing pipeline using OpenCV for input normalization and noise reduction.
- Trained, evaluated, and optimized deep learning models in TensorFlow; conducted systematic experimentation and model performance documentation.
- Integrated model inference through a FastAPI backend into a React-based web application, enabling real-time transcription — end-to-end AI system design.

PROJECTS

ModelSmith – LLM-Powered Autonomous AutoML System

Python · FastAPI · TensorFlow/Keras · Scikit-learn · Optuna · Docker

- Developed an intelligent AutoML framework leveraging LLM-driven dataset profiling and automatic model architecture selection — core agentic AI workflow design.
- Implemented competitive evaluation across Logistic Regression, Naive Bayes, k-NN, SVM, Random Forest, XGBoost, and Neural Networks with staged elimination strategy.
- Designed Optuna-based hyperparameter optimization pipeline, automated code generation, and secure Docker-based sandbox execution for robust model validation.

AI Doctor – Multi-Model Medical Prediction System

Python · TensorFlow · LSTM · CNN · FastAPI · React

- Built NLP-based disease prediction system using LSTM networks with full pipeline: tokenization, embeddings, and sequence modeling — directly applicable to RAG and semantic retrieval.
- Developed CNN-based liver disease prediction model and integrated both models into a FastAPI backend with React frontend for visualization.
- Demonstrates proficiency in building multi-model AI orchestration systems and scalable backend services.

Competitor Intelligence Agent – Autonomous Market Monitoring System

Python · LangChain · LangGraph · Playwright · NewsAPI · FastAPI · ChromaDB · RAG

- Built an autonomous multi-agent system that scrapes competitor websites using Playwright, detects week-on-week changes, and pulls latest news via NewsAPI for real-time market intelligence.
- Designed a LangGraph-orchestrated agentic pipeline with specialized agents: Scraper Agent, News Monitor Agent, Analysis Agent, and Report Generator Agent for end-to-end automation.
- Implemented RAG-powered analysis where an LLM reads all collected data, identifies patterns and competitor moves, and auto-generates a structured weekly PDF battle card with scheduled email delivery.

TECHNICAL SKILLS

Languages: Python (Expert), Java, C++, JavaScript, SQL

AI/ML & GenAI: LLMs, Prompt Engineering, Embeddings, RAG, CAG, Agentic AI, Scikit-learn, TensorFlow, PyTorch, CNN, LSTM, Vision Transformers

AI Frameworks: LangChain, LangGraph, AutoGen, Vector Databases (ChromaDB), Semantic Search, Optuna

Backend & APIs: FastAPI, REST APIs, Docker

Frontend: React.js, Next.js (basic), Streamlit

Data Processing: NumPy, Pandas, OpenCV, Unstructured.io

EDUCATION

B.Tech – Computer Science & Engineering (Core) | Vellore Institute of Technology, Bhopal

2023 – 2027